5.1.2 How far?

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Equilibrium		
(a)	use of the terms *mole fraction* and *partial pressure*	**See also 3.2.3 Chemical Equilibrium**.
(b)	calculation of quantities present at equilibrium, given appropriate data	*M0.2*
(c)	the techniques and procedures used to determine quantities present at equilibrium	**Not** for K_p. HSW4 Opportunities to carry out experimental and investigative work.
(d)	expressions for K_c and K_p for homogeneous and heterogeneous equilibria **(see also 3.2.3 f)**	*M0.2* Note: liquid and solid concentrations are constant and are omitted in heterogeneous K_c and K_p expressions.
(e)	calculations of K_c and K_p, or related quantities, including determination of units **(see also 3.2.3 f)**	*M0.0, M0.1, M0.2, M0.4, M2.2, M2.3, M2.4* Learners will **not** be required to solve quadratic equations.
(f)	**(i)** the qualitative effect on equilibrium constants of changing temperature for exothermic and endothermic reactions **(ii)** the constancy of equilibrium constants with changes in concentration, pressure or in the presence of a catalyst	*M0.3*
(g)	explanation of how an equilibrium constant controls the position of equilibrium on changing concentration, pressure and temperature	*M0.3*
(h)	application of the above principles in **5.1.2 How far?** for K_c, K_p to other equilibrium constants, where appropriate **(see also 5.1.3 c etc.)**.	